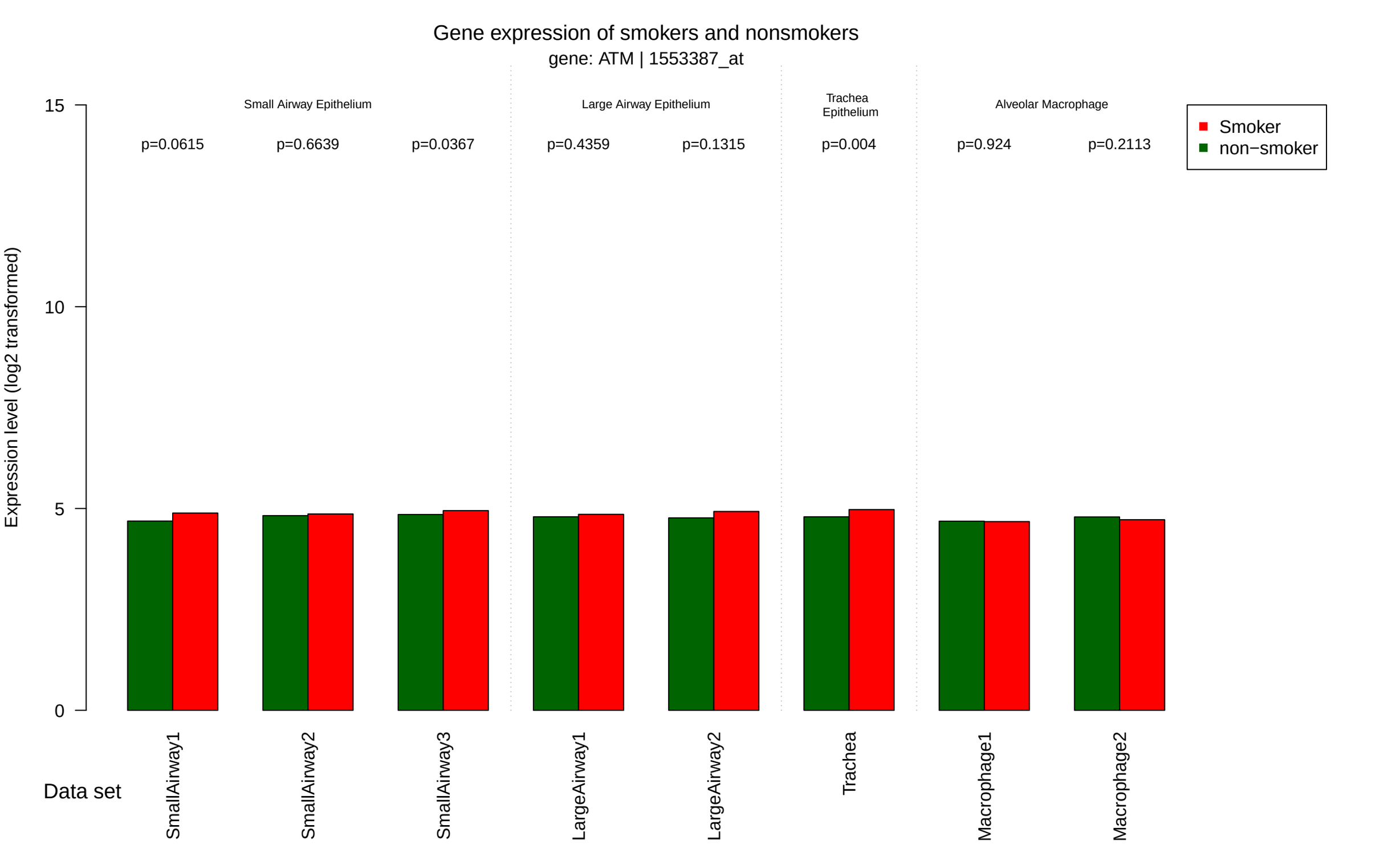
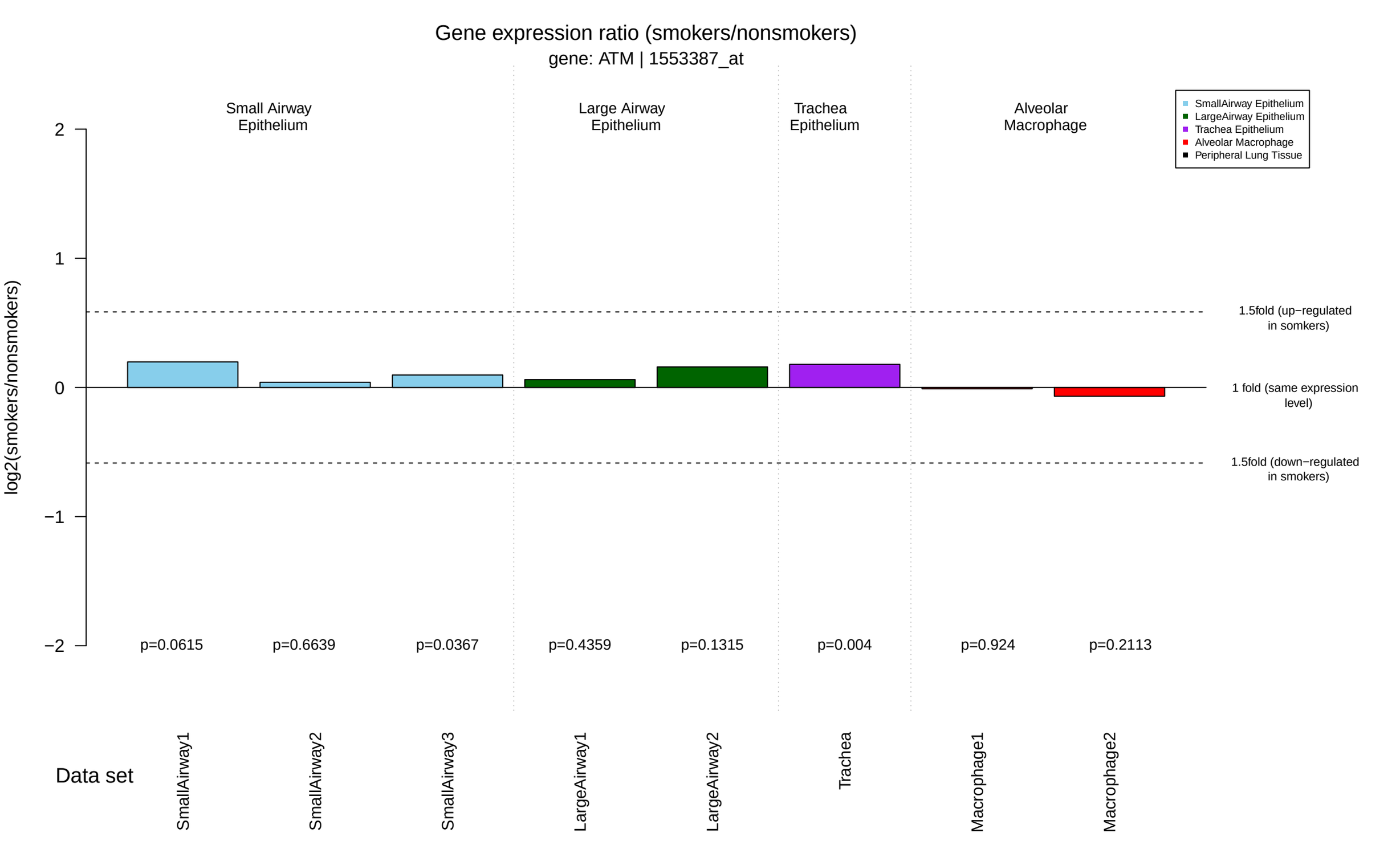
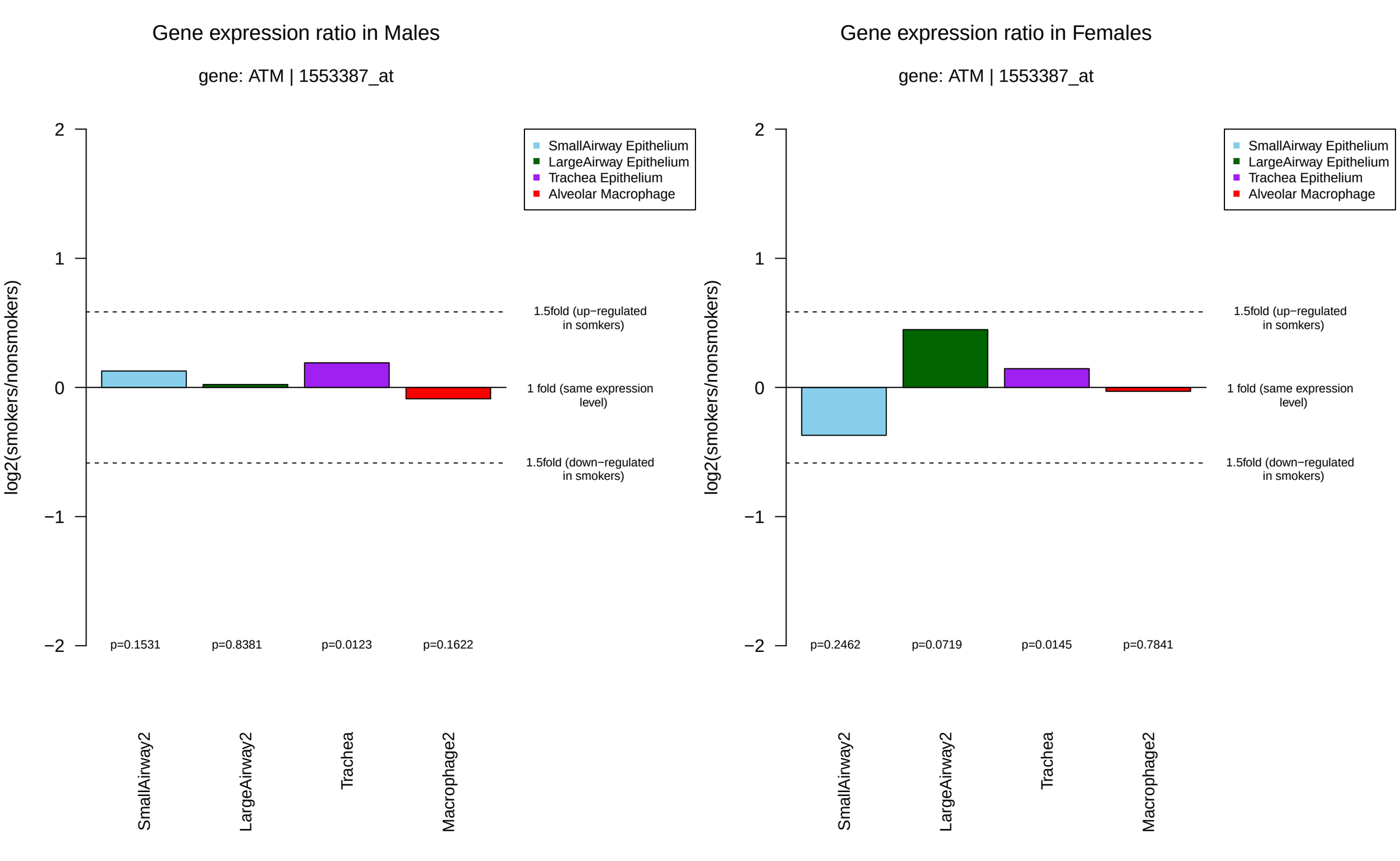
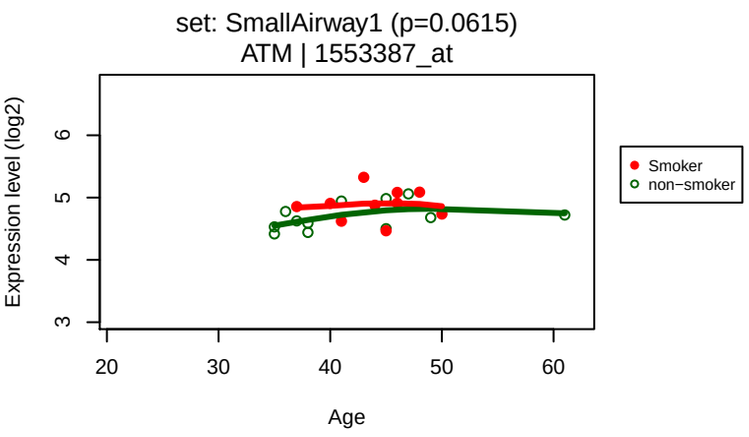
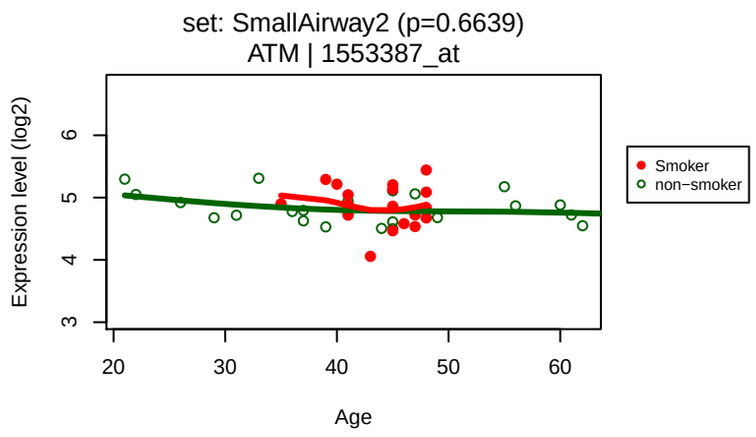
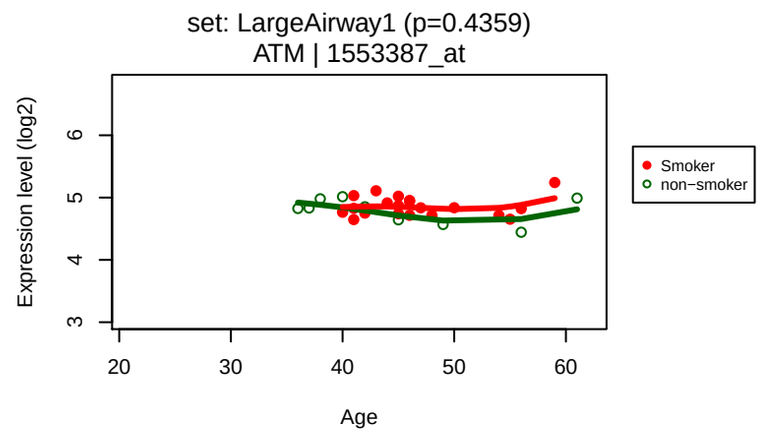

Gene expression of smokers and nonsmokers
gene: ATM | 1553387_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0615
p=0.6639
p=0.0367
p=0.4359
p=0.1315
p=0.004
p=0.924
p=0.2113
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

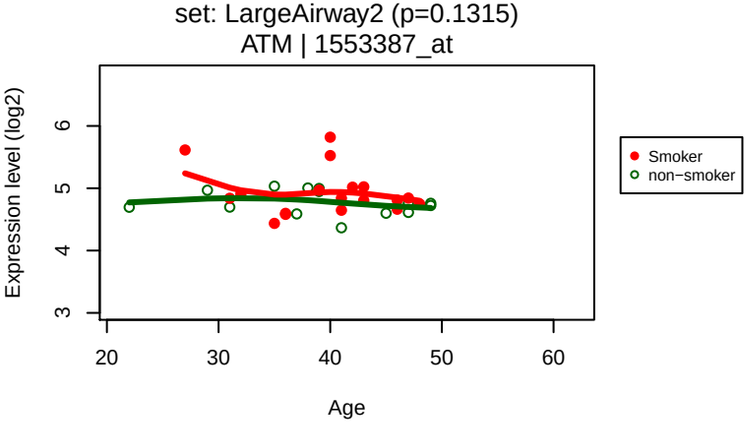

Gene expression ratio (smokers/nonsmokers)
gene: ATM | 1553387_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0615
p=0.6639
p=0.0367
p=0.4359
p=0.1315
p=0.004
p=0.924
p=0.2113
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

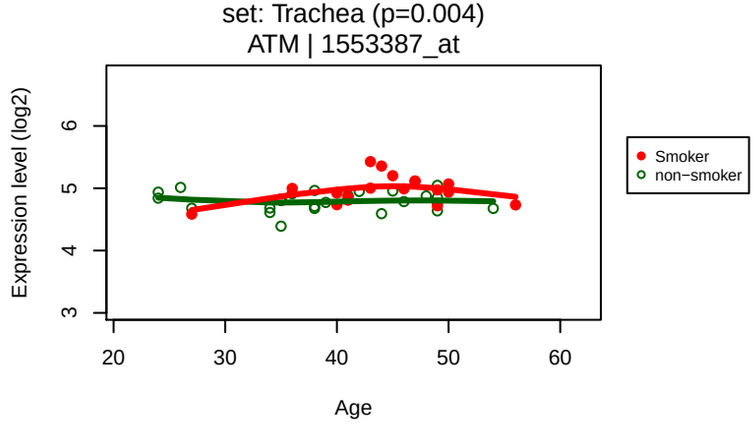

Gene expression ratio in Males
gene: ATM | 1553387_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1531
p=0.8381
p=0.0123
p=0.1622
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATM | 1553387_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2462
p=0.0719
p=0.0145
p=0.7841
SmallAirway2
LargeAirway2
Trachea
Macrophage2

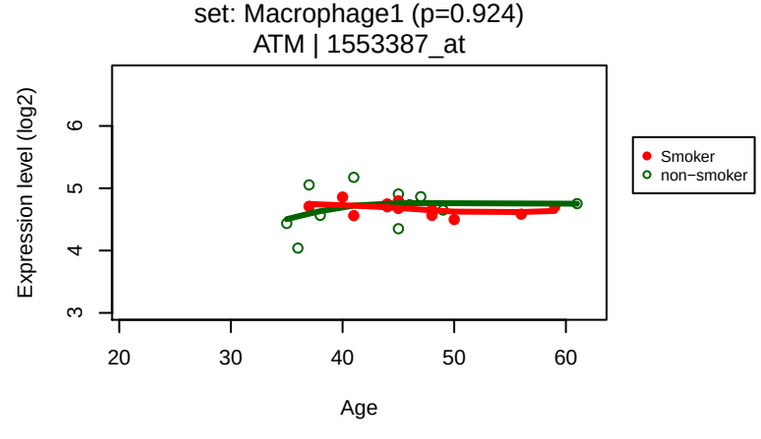

set: SmallAirway1 (p=0.0615)
ATM | 1553387_at
Expression level (log2)
Age
Smoker
non−smoker

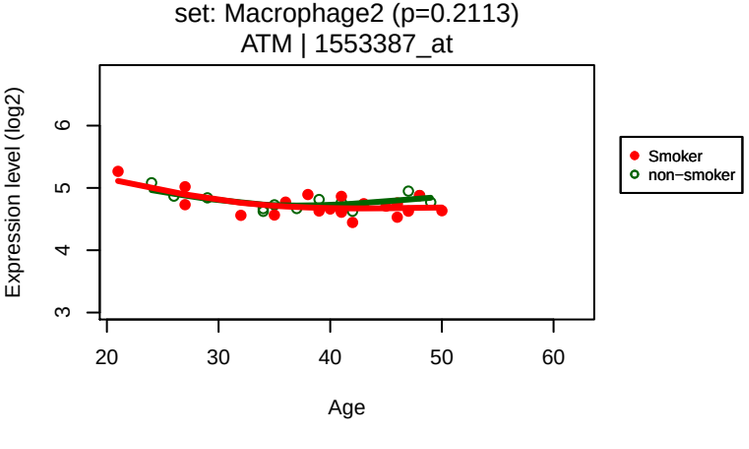

set: SmallAirway2 (p=0.6639)
ATM | 1553387_at
Expression level (log2)
Age
Smoker
non−smoker

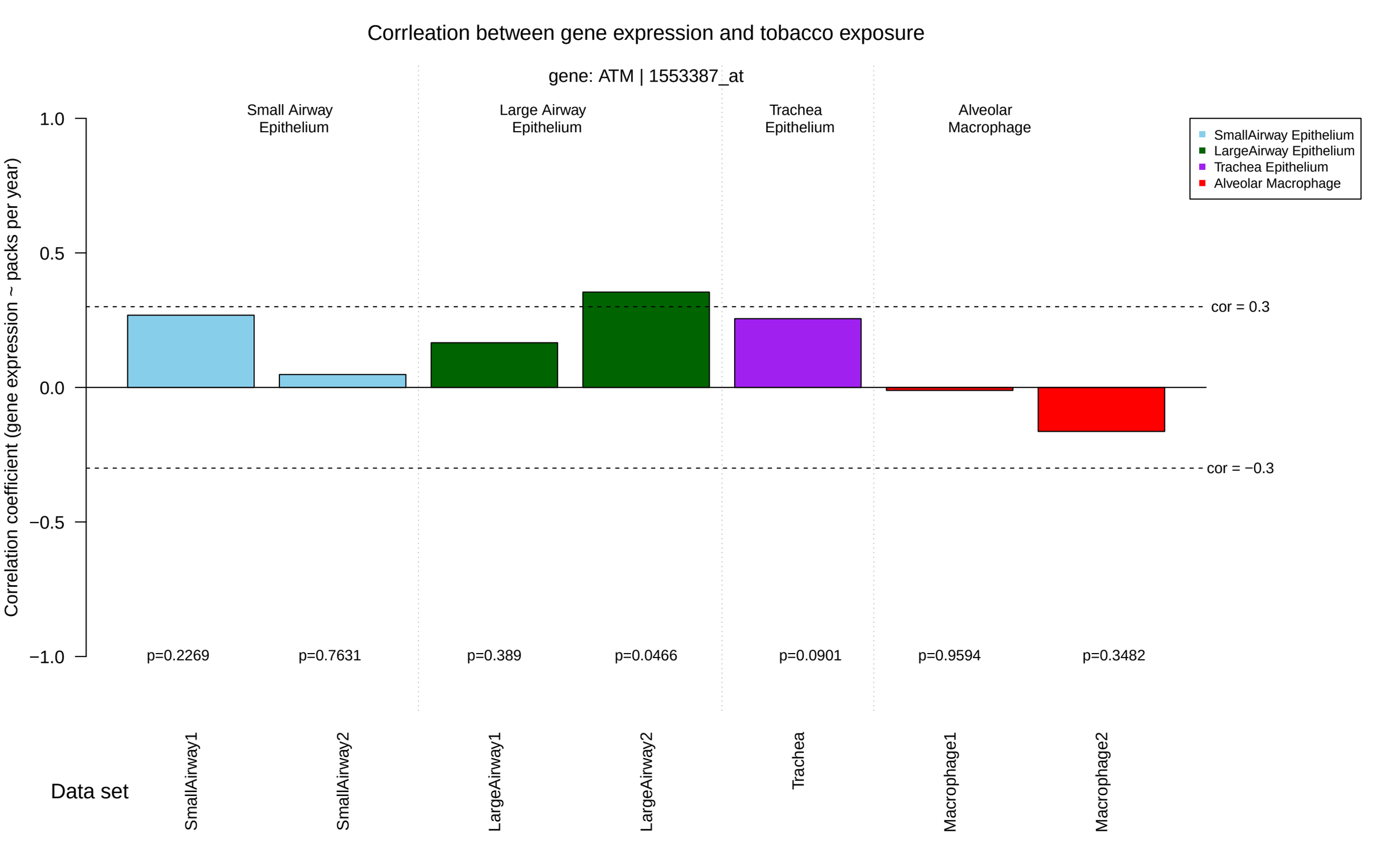

set: LargeAirway1 (p=0.4359)
ATM | 1553387_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1315)
ATM | 1553387_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.004)
ATM | 1553387_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.924)
ATM | 1553387_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2113)
ATM | 1553387_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATM | 1553387_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2269
p=0.7631
p=0.389
p=0.0466
p=0.0901
p=0.9594
p=0.3482
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2